

'crash test dummy', also referred to as 'Beater Box'. Test several or all of the changes on that system to see the effects.

Collecting hardware configuration data

There are several tools that you can use to collect important information about your machine hardware. This is very important in efficient performance tuning.

vmstat

vmstat provides information about processes, memory, paging, block I/O, traps, and CPU activity. It displays either average data or actual samples. You can enable the sampling mode when you additionally provide a sampling frequency and a sampling duration to vmstat.

The following command will display my machine's virtual memory report after a delay of two seconds, for four times.

```
# vmstat 2 4
```

Take a look at Figure 1 for the output of this command. Note that the first line of this report shows the averages since your last reboot. So there's no need to panic and you should ignore it.

vmstat displays the following statistics:

1. Process (procs) section
 - r – number of processes waiting for runtime
 - b – number of processes in uninterruptible sleep
2. Memory section
 - swpd – amount of virtual memory used (KB)
 - free – amount of idle memory (KB)
 - buff – amount of memory used as buffers (KB)
 - cache – amount of memory used as cache (KB)
3. Swap section
 - si – amount of memory swapped from the disk (KB per second)
 - so – Amount of memory

```
root@legacy ~]# vmstat 2 4
procs-----memory-----swap-----io-----system-----cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st
1 0 0 1977520 144720 831928 0 0 27 13 246 258 2 1 96 2 0
1 0 0 1977512 144720 831932 0 0 0 0 456 498 0 0 99 0 0
0 0 0 1977512 144720 831932 0 0 0 0 439 497 0 0 99 0 0
0 0 0 1977512 144720 831932 0 0 0 0 28 443 485 0 0 98 2 0
```

Figure 1: A typical vmstat output

- swapped to the disk (KB per second)
- 4. IO section
 - bi – blocks sent to a block device (blocks/s)
 - bo – blocks received from a block device (blocks/s)
- 5. System section
 - in – number of interrupts per second, including the clock
 - cs – number of context switches per second
- 6. CPU section
 - us – time spent running non-kernel code (user time, including nice time)
 - sy – time spent running kernel code (system time)
 - id – time spent idle

Asset Tag:

Handle 0x0200, DMI type 10, 6 bytes
On Board Device Information
Type: Video
Status: Enabled
Description: ATI MOBILITY Radeon X1400
Handle 0x0A01, DMI type 10, 6 bytes
On Board Device Information
Type: Sound
Status: Enabled
Description: Sigmatel 9200

In the same manner, you can get any information about your system. Check the man pages if you are not sure about the options. Running **dmidecode -t** will show you all the options that you can use:

dmidecode

The dmidecode command reads the system DMI table to display hardware and BIOS information of the system. This command will give you information on the current configuration of your system, as well as the system's maximum supported configuration. For example, **dmidecode** gives both the current RAM on the system and the maximum RAM supported by the system.

To get information about your motherboard, I can use the following command:

```
# dmidecode -t baseboard
```

Given below is the output of this command on my system:

```
Handle 0x0200, DMI type 2, 9 bytes
Base Board Information
Manufacturer: Dell Inc.
Product Name: 0XD720
Version:
Serial Number: BYN0C31
CN486436/AH147
```

dmidecode option requires an argument — 't'

Type number of keyword expected

Valid type keywords are:

```
birod
system
baseboard
chassis
processor
memory
cache
connector
slot
```

By using **dmidecode**, any of these options will give you detailed information about it. For instance, if I want to know about the CPU, I can now easily run: **dmidecode -t processor**

I can also use the **grep** with **dmidecode** to check how much RAM my system will support, as follows:

```
# dmidecode -t memory | grep -i Maximum
Maximum Capacity: 4GB
```

So this is a very handy tool